

Error estimates for elliptic obstacle FEM

Load approximation, truncation error, and iterative error

The obstacle is identically zero. This makes the discrete admissible set a subset of the continuous one and yields a computable truncation bound using the coefficients, Poincaré constant, interpolation constants, and $\|f_h\|_0$. Load approximation and iterative error are treated separately. Rounding error is not included.

1 The obstacle problem

Let $\Omega \subset \mathbb{R}^2$ be a bounded *convex polygon*, and let $f \in C(\overline{\Omega})$. The obstacle is $\psi \equiv 0$. Define

$$K = \{v \in H_0^1(\Omega) : v \geq 0 \text{ a.e. in } \Omega\}.$$

This is a nonempty closed convex set containing 0. Find $u \in K$ such that

$$a(u, v - u) \geq (f, v - u)_0 \quad (v \in K). \quad (\text{P})$$

Here the coefficients are constant, D is symmetric positive definite, $b \in \mathbb{R}^2$, $r \in \mathbb{R}$, and

$$\begin{aligned} Lv &= -\nabla \cdot (D \nabla v) + b \cdot \nabla v + rv, \\ a(w, v) &= \int_{\Omega} ((D \nabla w) \cdot \nabla v + (b \cdot \nabla w)v + r w v) \, dx. \end{aligned}$$

Write $\lambda = \lambda_{\min}(D) > 0$ and $\Lambda = \lambda_{\max}(D)$. Convexity is needed for the global H^2 estimate used below; a general nonconvex polygon requires an additional regularity assumption.

2 Notation and stability constants

We use $(w, v)_0 = \int_{\Omega} wv \, dx$ and

$$\|v\|_0^2 = \int_{\Omega} |v|^2 \, dx, \quad |v|_1^2 = \int_{\Omega} |\nabla v|^2 \, dx, \quad |v|_2^2 = \sum_{i,j=1}^2 \int_{\Omega} |\partial_{ij} v|^2 \, dx.$$

The second-order seminorm counts the mixed derivative twice. A subscript T restricts the integral to an element. Choose

$$\|v\|_0 \leq C_P |v|_1 \quad (v \in H_0^1(\Omega)), \quad \theta := \lambda - C_P^2 \max\{-r, 0\} > 0, \quad M := \Lambda + |b|C_P + |r|C_P^2. \quad (1)$$

One may take $C_P = \text{diam}(\Omega)/(\sqrt{2}\pi)$, by zero extension to an enclosing rectangle whose side lengths do not exceed $\text{diam}(\Omega)$. Since b is constant, $\int_{\Omega} (b \cdot \nabla v)v \, dx = 0$ for $v \in H_0^1(\Omega)$. Thus

$$a(v, v) \geq \theta |v|_1^2, \quad |a(w, v)| \leq M |w|_1 |v|_1. \quad (2)$$

These inequalities imply existence and uniqueness of the variational inequalities considered here.

3 Finite element approximation and interpolation

Let $\mathcal{T}_h$ be a conforming shape-regular triangulation, with $h = \max_{T \in \mathcal{T}_h} \text{diam}(T)$. Define

$$S_h = \{v \in C^0(\bar{\Omega}) : v|_T \in \mathbb{P}_1(T)\}, \quad V_h = S_h \cap H_0^1(\Omega).$$

Let $f_h = \mathcal{I}_h f \in S_h$ be the nodal interpolant of f , using all nodes, including boundary nodes. Define

$$K_h = \{v_h \in V_h : v_h \geq 0 \text{ in } \Omega\} = V_h \cap K.$$

For piecewise affine functions, nonnegativity is equivalent to nonnegative nodal values. In particular, $K_h \subset K$. The exact FE solution $u_h \in K_h$ satisfies

$$a(u_h, v_h - u_h) \geq (f_h, v_h - u_h)_0 \quad (v_h \in K_h). \quad (\text{FEM})$$

For $k = 0, 1$, let

$$I_k = \max_{T \in \mathcal{T}_h} \sup_{\substack{z \in H^2(T), \\ z=0 \text{ at the vertices of } T \\ |z|_{2,T} > 0}} \frac{|z|_{k,T}}{|z|_{2,T}}, \quad |z|_{0,T} := \|z\|_{L^2(T)}.$$

Since $z - \mathcal{I}_h z$ vanishes at element vertices and has the same Hessian as z on each element,

$$\|z - \mathcal{I}_h z\|_0 \leq I_0 |z|_2, \quad |z - \mathcal{I}_h z|_1 \leq I_1 |z|_2 \quad (z \in H^2(\Omega)). \quad (3)$$

Valid computable upper bounds for I_0, I_1 may replace their exact values. On a shape-regular mesh family, $I_0 = O(h^2)$ and $I_1 = O(h)$.

4 Error analysis

Introduce $\tilde{u} \in K$, with the zero obstacle and interpolated load:

$$a(\tilde{u}, v - \tilde{u}) \geq (f_h, v - \tilde{u})_0 \quad (v \in K). \quad (\text{P}')$$

Set $\delta = u - \tilde{u}$ and $e = \tilde{u} - u_h$, so $u - u_h = \delta + e$.

4.1 Original problem versus auxiliary problem

Take $v = \tilde{u}$ in (P) and $v = u$ in (P'). Adding the inequalities gives

$$a(u, \tilde{u} - u) + a(\tilde{u}, u - \tilde{u}) \geq (f - f_h, \tilde{u} - u)_0,$$

or $a(\delta, \delta) \leq (f - f_h, \delta)_0$. Hence

$$\theta |\delta|_1^2 \leq \|f - f_h\|_0 \|\delta\|_0 \leq C_P \|f - f_h\|_0 |\delta|_1.$$

After cancellation (with the zero case immediate),

$$\boxed{|u - \tilde{u}|_1 \leq \frac{C_P}{\theta} \|f - f_h\|_0, \quad \|u - \tilde{u}\|_0 \leq \frac{C_P^2}{\theta} \|f - f_h\|_0.} \quad (4)$$

Only continuity of f is assumed. If additionally $f \in H^2(\Omega)$, then $\|f - f_h\|_0 \leq I_0 |f|_2$; otherwise that interpolation estimate is not asserted for f .

4.2 Auxiliary problem versus the FE problem: Falk's estimate

The regularity established in Sections 4.3–4.4 gives $\tilde{u} \in H^2(\Omega)$ and $q := L\tilde{u} - f_h \in L^2(\Omega)$. For arbitrary $v_h \in K_h$ and $v \in K$, use (FEM) to obtain

$$\begin{aligned} \theta |e|_1^2 &\leq a(e, e) = a(e, \tilde{u} - v_h) + a(e, v_h - u_h) \\ &\leq a(e, \tilde{u} - v_h) + (q, v_h - u_h)_0. \end{aligned}$$

Here $a(\tilde{u}, z) = (L\tilde{u}, z)_0$ for $z \in H_0^1(\Omega)$. Decompose

$$v_h - u_h = (v_h - \tilde{u}) + (\tilde{u} - v) + (v - u_h).$$

The auxiliary inequality says $(q, v - \tilde{u})_0 \geq 0$, so

$$\begin{aligned} \theta |e|_1^2 &\leq M |e|_1 |\tilde{u} - v_h|_1 + \|q\|_0 (\|\tilde{u} - v_h\|_0 + \|v - u_h\|_0) \\ &\leq \frac{\theta}{2} |e|_1^2 + \frac{M^2}{2\theta} |\tilde{u} - v_h|_1^2 + \|q\|_0 (\|\tilde{u} - v_h\|_0 + \|v - u_h\|_0). \end{aligned}$$

Thus the required version of Falk's estimate is

$$\frac{\theta}{2} |\tilde{u} - u_h|_1^2 \leq \|q\|_0 (\|\tilde{u} - v_h\|_0 + \|v - u_h\|_0) + \frac{M^2}{2\theta} |\tilde{u} - v_h|_1^2. \quad (5)$$

The argument uses coercivity and continuity, not symmetry of a .

Comparison functions for the zero obstacle

Choose $v_h = \mathcal{I}_h \tilde{u}$ and $v = u_h$. Regularity gives continuity of $\tilde{u}$; since $\tilde{u} \geq 0$, its nodal interpolant belongs to K_h . Also $u_h \in K$ because $K_h \subset K$. Therefore $\|v - u_h\|_0 = 0$, while

$$\|\tilde{u} - \mathcal{I}_h \tilde{u}\|_0 \leq I_0 |\tilde{u}|_2, \quad |\tilde{u} - \mathcal{I}_h \tilde{u}|_1 \leq I_1 |\tilde{u}|_2.$$

Substitution in (5) yields

$$\boxed{\frac{\theta}{2} |\tilde{u} - u_h|_1^2 \leq I_0 \|f_h - L\tilde{u}\|_0 |\tilde{u}|_2 + \frac{M^2 I_1^2}{2\theta} |\tilde{u}|_2^2.} \quad (6)$$

The obstacle-interpolation term from the general-obstacle problem has disappeared. The remaining task is to bound both $|\tilde{u}|_2$ and $\|f_h - L\tilde{u}\|_0$ using $\|f_h\|_0$.

4.3 Bounds for $L\tilde{u}$ and the residual

Write $F_h = \|f_h\|_0$. Regularity theory of variational inequalities [1] implies $L\tilde{u} \in L^2(\Omega)$ and $\|L\tilde{u}\|_0 \leq F_h$. Elliptic regularity on the convex polygon then yields $\tilde{u} \in H^2(\Omega) \cap H_0^1(\Omega)$.

There is an explicit identification of $L\tilde{u}$. Let $\mathcal{C} = \{x \in \Omega : \tilde{u}(x) = 0\}$ and $\mathcal{N} = \{x \in \Omega : \tilde{u}(x) > 0\}$. Stampacchia's level-set lemma states that the weak gradient of a Sobolev function vanishes almost everywhere on each of its level sets. Applied first to $\tilde{u}$, it gives $\nabla \tilde{u} = 0$ a.e. on $\mathcal{C}$. Applied next to each $\partial_i \tilde{u} \in H^1(\Omega)$, it gives $\partial_{ji} \tilde{u} = 0$ a.e. on $\mathcal{C}$. Thus $L\tilde{u} = 0$ there.

The set $\mathcal{N}$ is open. For every $\eta \in C_c^\infty(\mathcal{N})$, both $\tilde{u} + t\eta$ and $\tilde{u} - t\eta$ are admissible for sufficiently small $t > 0$. Testing the auxiliary inequality in both directions gives $a(\tilde{u}, \eta) = (f_h, \eta)_0$, so $L\tilde{u} = f_h$ a.e. on $\mathcal{N}$. Consequently,

$$L\tilde{u} = f_h \chi_{\mathcal{N}}, \quad f_h - L\tilde{u} = f_h \chi_{\mathcal{C}}, \quad \boxed{\|L\tilde{u}\|_0 \leq F_h, \quad \|f_h - L\tilde{u}\|_0 \leq F_h.} \quad (7)$$

Here χ_E denotes the indicator of E . Neither the contact set nor the auxiliary solution must be computed to use these bounds.

4.4 An estimate of $\tilde{u}$ in H^2

For the zero-Dirichlet solution z of $Lz = g \in L^2(\Omega)$,

$$|z|_1 \leq \frac{C_P}{\theta} \|g\|_0, \quad \|z\|_0 \leq \frac{C_P^2}{\theta} \|g\|_0.$$

The change of variables $x = D^{1/2}y$ preserves convexity. Applying Miranda–Talenti on the transformed domain gives $|z|_2 \leq \lambda^{-1} \|\nabla \cdot (D\nabla z)\|_0$. It follows that

$$|z|_2 \leq \frac{1}{\lambda} (\|g\|_0 + |b| |z|_1 + |r| \|z\|_0) \leq R \|g\|_0, \quad R := \frac{1}{\lambda} \left(1 + \frac{|b|C_P}{\theta} + \frac{|r|C_P^2}{\theta} \right).$$

Taking $g = L\tilde{u}$ gives

$$\boxed{|\tilde{u}|_2 \leq R \|L\tilde{u}\|_0 \leq RF_h.} \quad (8)$$

4.5 The computable truncation estimate

Substituting (7) and (8) into (6) gives

$$\frac{\theta}{2} |\tilde{u} - u_h|_1^2 \leq RI_0 F_h^2 + \frac{M^2 R^2 I_1^2}{2\theta} F_h^2.$$

Therefore

$$|\tilde{u} - u_h|_1 \leq E_H := \left(\frac{2R}{\theta} I_0 + \frac{M^2 R^2}{\theta^2} I_1^2 \right)^{1/2} \|f_h\|_0. \quad (9)$$

Poincaré supplies the corresponding L^2 bound:

$$\|\tilde{u} - u_h\|_0 \leq E_L := C_P E_H. \quad (10)$$

Given the coefficients, these use only C_P , I_0, I_1 (or valid computable upper bounds), and $\|f_h\|_0$. No unknown obstacle seminorm or contact-set quantity remains. For $f_h = \sum_i c_i \phi_i$ in the full nodal basis of S_h ,

$$\|f_h\|_0^2 = c^\top M c, \quad M_{ij} = \int_\Omega \phi_i \phi_j \, dx.$$

The bound need not be sharp. For fixed data on a shape-regular family, it gives $E_H = O(h)$ and $E_L = O(h)$.

The original load and a general-obstacle transformation

The complete error for the original load still includes $\|f - f_h\|_0$:

$$|u - u_h|_1 \leq \frac{C_P}{\theta} \|f - f_h\|_0 + E_H, \quad (11)$$

$$\|u - u_h\|_0 \leq \frac{C_P^2}{\theta} \|f - f_h\|_0 + E_L. \quad (12)$$

Computing E_H, E_L does not evaluate the separate error due to replacing f by f_h .

The bounds above concern the exact FE solution. The next section adds the error from terminating the iteration; rounding error remains outside the analysis.

5 Iterative error analysis

5.1 Nodal form and the projection equation

Let $\{\varphi_i\}_{i=1}^N$ be the basis of V_h associated with the interior nodes, and write $u_h = \sum_j U_j^* \varphi_j$. Define

$$A_{ij} = a(\varphi_j, \varphi_i), \quad F_i = (f_h, \varphi_i)_0, \quad \mathcal{B} = [0, \infty)^N.$$

The vector F is the assembled load vector, not the nodal values of f_h . The discrete inequality is

$$(AU^* - F)^\top (V - U^*) \geq 0 \quad (V \in \mathcal{B}). \quad (13)$$

For every $\rho > 0$, the characterization of Euclidean projection gives

$$U^* = G(U^*), \quad G(U) = P_{\mathcal{B}}(U + \rho(F - AU)), \quad (P_{\mathcal{B}}Z)_i = \max\{0, Z_i\}. \quad (14)$$

Indeed, $X = P_{\mathcal{B}}Z$ exactly when $(Z - X)^\top (V - X) \leq 0$ for every $V \in \mathcal{B}$.

5.2 A contraction constant

All vector norms in this section are Euclidean. Let

$$S = \frac{A + A^\top}{2}, \quad 0 < \alpha \leq \lambda_{\min}(S), \quad \beta \geq \|A\|_2. \quad (15)$$

Coercivity ensures S is positive definite. In particular, $Z^\top AZ \geq \alpha\|Z\|_2^2$ and $\|AZ\|_2 \leq \beta\|Z\|_2$. Nonexpansiveness of the projection gives, with $Z = U - V$,

$$\begin{aligned} \|G(U) - G(V)\|_2^2 &\leq \|(I - \rho A)Z\|_2^2 \\ &= \|Z\|_2^2 - 2\rho Z^\top AZ + \rho^2 \|AZ\|_2^2 \\ &\leq (1 - 2\rho\alpha + \rho^2\beta^2)\|Z\|_2^2. \end{aligned}$$

Consequently, with

$$\boxed{\rho = \frac{\alpha}{\beta^2}, \quad C = \sqrt{1 - \frac{\alpha^2}{\beta^2}} < 1, \quad \|G(U) - G(V)\|_2 \leq C\|U - V\|_2,} \quad (16)$$

the iteration $U^{k+1} = G(U^k)$ converges to U^* .

Correction for convection. If A is symmetric positive definite, α and β may be its smallest and largest singular values, as in the handwritten formula. For nonsymmetric A , the smallest singular value does not provide the required lower bound on $Z^\top AZ$; use the symmetric part as in (15). Valid numerical bounds for α, β are prerequisites for a numerical contraction certificate; their verification is not supplied by this note.

5.3 A posteriori iterative error and stopping rule

For $0 < C < 1$, the geometric-series estimate gives

$$\begin{aligned} \|U^* - U^{k+1}\|_2 &\leq \sum_{j=k+1}^{\infty} \|U^{j+1} - U^j\|_2 \\ &\leq \sum_{\ell=1}^{\infty} C^\ell \|U^{k+1} - U^k\|_2 = \frac{C}{1-C} \|U^{k+1} - U^k\|_2. \end{aligned}$$

Thus, for a prescribed Euclidean tolerance $\varepsilon > 0$,

$$\boxed{\|U^{k+1} - U^k\|_2 \leq \varepsilon \frac{1-C}{C} \implies \|U^{k+1} - U^*\|_2 \leq \varepsilon.} \quad (17)$$

The guarantee applies to the *newer* iterate U^{k+1} . A corresponding implementation in exact arithmetic is:

```
U0 = initial guess
U1 = G(U0)
while norm(U1 - U0) > epsilon*(1-C)/C:
    U0 = U1
    U1 = G(U1)
return U1
```

If $C = 0$, G is constant and a single update gives U^* ; the stopping formula involving division by C is then unnecessary.

5.4 Function norms, error separation, and rounding

The tolerance in (17) is a coefficient-vector tolerance. To relate it to the FE function $u_h^{k+1} = \sum_i U_i^{k+1} \varphi_i$, define

$$H_{ij} = \int_{\Omega} \nabla \varphi_j \cdot \nabla \varphi_i \, dx, \quad B_{ij} = \int_{\Omega} \varphi_j \varphi_i \, dx.$$

Writing $d = U^{k+1} - U^*$, one has

$$\left| u_h^{k+1} - u_h \right|_1^2 = d^T H d, \quad \left\| u_h^{k+1} - u_h \right\|_0^2 = d^T B d.$$

If upper bounds $\gamma_H \geq \sqrt{\lambda_{\max}(H)}$ and $\gamma_L \geq \sqrt{\lambda_{\max}(B)}$ are supplied, the iterative contributions are at most $\gamma_H \varepsilon$ and $\gamma_L \varepsilon$. These are additional conditional matrix bounds. The complete separation is

$$u - u_h^{k+1} = (u - \tilde{u}) + (\tilde{u} - u_h) + (u_h - u_h^{k+1}).$$

Accordingly, the bounds in (11)–(12) gain the terms $\gamma_H \varepsilon$ and $\gamma_L \varepsilon$, respectively.

6 Connection with the program

The program evaluates an upper bound for the error

$$|\tilde{u} - u_h^{k+1}|_1, \quad (18)$$

between the auxiliary obstacle problem P' and the result of the iterative algorithm in section 5.3.

We take $\hat{I}_0 \geq I_0$, $\hat{I}_1 \geq I_1$ as computable interpolation bound obtained from the mesh, with explicit formula given in [2]. Then 9 yields

$$\boxed{|\tilde{u} - u_h|_1 \leq E_H := \left(\frac{2R}{\theta} \hat{I}_0 + \frac{M^2 R^2}{\theta^2} \hat{I}_1^2 \right)^{1/2} \|f_h\|_0.} \quad (19)$$

We evaluate the largest eigenvalue Λ of the elastic matrix H given in section 5.4, then we have

$$\boxed{|\tilde{u} - u_h^{k+1}|_1 \leq \Lambda^{1/2} \epsilon.} \quad (20)$$

Rounding error has not been addressed. The contraction and stopping proof assume exact assembly and exact evaluation of G and the norms. Floating-point errors in assembly, iteration, and evaluation of the error bounds are separate from the iterative error estimated above. No rounding-error-certified bound is claimed.

References

- [1] H. Brezis and G. Stampacchia, “Sur la régularité de la solution d’inéquations elliptiques,” *Bulletin de la Société Mathématique de France*, vol. 96, pp. 153–180, 1968. [Online]. Available: <https://api.semanticscholar.org/CorpusID:123884479>
- [2] K. Kobayashi, “Remarkable upper bounds for the interpolation error constants on the triangles,” 2025. [Online]. Available: <https://arxiv.org/abs/2507.04032>